

# THUY LINH DO

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Chicago, IL - 60637, USA

## EDUCATION

- **Tulane University** Jul 2019 - May 2025  
*Doctor of Philosophy - PhD in Mathematics*  
◦ GPA: 4.00/4.00  
New Orleans, LA, USA
- **Vietnam National University - University of Education** Sep 2013 - Jul 2017  
*Bachelor in Mathematics Teacher Education, Valedictorian*  
◦ GPA: 3.81/4.00  
Hanoi, Vietnam

## EMPLOYMENT HISTORY

- **Tulane University** 2025 - present  
*Research Scientist*  
New Orleans, LA, USA
- **Tulane University** 2019 - 2025  
*PhD Candidate/Graduate Teaching Associate/Graduate Research Assistant*  
New Orleans, LA, USA
- **NSF-Simons Southeast Center for Mathematics and Biology** 2022 - 2025  
*Junior Researcher*  
Atlanta, GA, USA
- **Vietnam National University, University of Science** 2017 - 2019  
*Lecturer*  
Hanoi, Vietnam

## RESEARCH INTERESTS

Mathematical and Statistical Modeling, Data Analysis, Change-Point Detection, Intracellular Transport, Machine Learning, Hierarchical Models, Bayesian Inference.

## TEACHING TRAINING AND EXPERIENCE

- **Graduate Teaching Assistant** 2019 - 2025  
*Tulane University - Department of Mathematics*
  - MATH 1110: Probability and Statistics
  - MATH 1230: Statistics for Scientists
  - MATH 1210: Calculus I
  - MATH 1220: Calculus II
  - MATH 2210: Calculus III
- **Lecturer** 2017-2019  
*Vietnam National University, University of Science*
  - Applied Statistics
  - Mathematical Statistics and Probability
  - Calculus I, II, III
  - Linear Algebra
- **Part-time Teacher** 2017-2019  
*High School for Gifted Students, VNU University of Science*
  - Pre-Calculus
  - Pre-Algebra
  - Probability and Statistics
- **B.S in Education with a six-month internship** 2016-2017  
*High School of Education and Sciences, VNU University of Education*
  - Calculus
  - Analysis
  - Probability and Statistics

## SKILLS

- **Programming:** Python, R, LaTeX, Maple, SQL, MatLab, Microsoft Office.
- **Python:** Influential in using JAX, tskit, Numpy, Matplotlib, PyTorch
- **R:** Influential in using RStudio, including R Script, R Markdown, packages ggplot2, tidyverse, dplyr, stats, MASS, caret, tidymodels, parallel
- **Languages:** Strong reading, writing and speaking competencies for English and Vietnamese.

## RESEARCH PUBLICATIONS

I= IN PREPARATION, B= BOOKS AND CHAPTERS, J=JOURNAL, S=IN SUBMISSION, T=THESIS

- [T] **Linh Do (2025). *Changepoint detection: theory, algorithms, and application in intracellular transport*. Tulane University.**
- [J.1] **Dat Do, Linh Do, XuanLong Nguyen (2025). *Strong identifiability and parameter learning in regression with heterogeneous response*. *Electronic Journal of Statistics*, 19 (1) 131 - 203.**
- [S.1] **Dat Do, Linh Do, Scott A. McKinley, XuanLong Nguyen, Jonathan Terhorst (2024). *Dendrogram of mixing measures: Hierarchical Clustering and model selection for finite mixture models*. The manuscript is under review with *Journal of the American Statistical Association*.**
- [S.2] **Keisha J. Cook, Nathan Rayens, Linh Do, Christine K. Payne, and Scott A. McKinley (2024). *Considering experimental frame rates and robust segmentation analysis of piecewise-linear microparticle trajectories*. The manuscript is accepted for publication in *Mathematical Biosciences and Engineering*.**
- [B.1] **Nguyen, A.T., Pham, H.T.T, Trinh, Q.A, Linh Do, Dang, P.A, & Hens, L., (2021). *The geography of climate change adaptation in the vietnam northern mountains: A quantitative analysis for intentions of indigenous ethnic minorities using structural equation modeling (sem) and protection motivation theory (pmt)*. In: Rai, P.K., Singh, P., Mishra, V.N. (eds) *Recent Technologies for Disaster Management and Risk Reduction*. Earth and Environmental Sciences Library. Springer, Cham.**
- [I.1] **Linh Do and et al. *Change-in-velocity detection for multidimensional data*. To be submitted to *Journal of Computational and Graphical Statistics***
- [I.2] **Linh Do and et al. *Dendrogram of signal functions for summarization and model selection in changepoint detection*. In preparation for submission to the *Journal of Statistical Planning and Inference*.**

## PRESENTATIONS

- **Strong identifiability and parameter learning in regression with heterogeneous response.** 13th International Conference on Bayesian Nonparametrics (Puerto Varas, Chile). Oct 2022.
- **Mixture Model and the applications.** Graduate Colloquium at the Department of Mathematics, Tulane University. Nov 2022.
- **A new changepoint algorithm detecting multiple change-in-velocity.** NSF-Simons Southeast Center for Mathematics and Biology (SCMB). Oct 2023.
- **Dendrogram of mixing measures: Hierarchical clustering and model selection for finite mixture models.** Joint Statistical Meetings (Portland, Oregon). Aug 2024
- **Inferring and interpreting heterogeneous data using dendrograms.** Computational and Applied Math Seminar at the Department of Mathematics, Tulane University. Nov 2024.
- **CPLASS: A segmentation analysis in noisy trajectories.** Final Symposium, NSF-Simons Southeast Center for Mathematics and Biology. Apr 2025.

## HONORS AND AWARDS

- **Mathematical Sciences and Research Institute (MSRI) Summer Program Admission** 2022  
*MSRI & Courant Institute in New York*
  - The award was being selected into the program. Mathematics of Machine Learning (INdAM and Courant Institute).
- **Toshiba Scholarship for young scientists** 2018  
*Toshiba*
  - The Toshiba scholarship program aims to support students, graduate students, and graduate students (students) with excellent academic and research achievements.
- **Golden record of excellent valedictorians graduating from universities in Hanoi, Vietnam** 2017  
*Hanoi Municipal Party Committee - Hanoi People's Council - Hanoi People's Committee.*
  - The program to honor outstanding valedictorians graduating from universities and academies in Hanoi in 2017, organized by the Hanoi Municipal Party Committee, the People's Council, and the People's Committee.
- **Prize for Outstanding Student Performance** 2017  
*Hanoi University of Science and Hanoi University of Education, VNU*
  - For students with excellent achievement in learning and scientific research.

## VOLUNTEER EXPERIENCE

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- **Conference Organizer, Marketing and Photography** 2019  
*Math For All - Tulane University - New Orleans - Louisiana.*
- **Session Leader** 2020  
*Boys at Tulane in STEM (BATS) - K-12 STEM Education Events - Tulane University - New Orleans - Louisiana.*

## REFERENCES

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1. **Scott A. McKinley**  
Associate Professor, Department of Mathematics  
Tulane University  
Email: scott.mckinley@tulane.edu  
Phone: +1-504-862-3246  
*Role: Thesis Advisor / Chair of Dissertation Committee*
2. **Lifeng Han**  
Professor of Practice, Department of Mathematics  
Tulane University  
Email: lhan1@tulane.edu  
Phone: +1-504-862-3435  
*Role: Professor*
3. **Michael Joyce**  
Senior Professor of Practice, Department Chair, Department of Mathematics  
Tulane University  
Email: mjoyce3@tulane.edu  
Phone: +1-504-862-3438  
*Role: Advisor/ Professor*
4. **XuanLong Nguyen**  
Professor, Department of Statistics  
University of Michigan  
Email: xuanlong@umich.edu  
Phone: +1-734-763-3499  
*Role: Professor*
5. **Viet Cuong Do**  
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Vietnam National University, Hanoi University of Science  
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Phone: +84 868961764  
*Role: Professor*